FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – V
PAPER –1
TEST DATE: 08-02-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A ***(01 –06, 19 – 24, 37 – 42):*** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A ***(07 – 10, 25 – 28, 43 – 46):*** This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B ***(11 – 18, 29 – 36, 47 – 54):*** This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

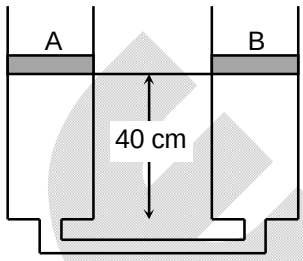
Physics

PART – I

Section – A (Maximum Marks: 24)

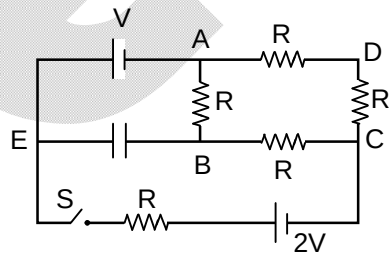
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. Two vertical conducting cylinders A and B of different cross sections are connected by a thin tube as shown. An ideal gas is trapped in the cylinders by two airtight pistons of masses 1.0 kg and 2.0 kg respectively. Initially the pistons are at the same height 40cm above the base of the cylinders. If an additional mass of 10 g were gently placed on the piston in the cylinder A, which of the following statements correctly describe the new steady state ultimately reached.



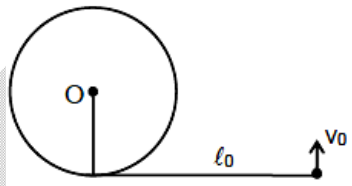
(A) Pressure of the gas has been increased.
(B) Pressure of the gas remains unchanged.
(C) Height difference between the pistons becomes 30 cm.
(D) Height difference between the pistons becomes 60 cm.

2. Five equal resistances, a capacitor and two batteries are connected in a circuit as shown in the figure. The capacitor is initially uncharged and each battery has negligible internal resistance. It is given that R = 1Ω, C = 1μF and V = 8 volt. On the basis of above information answer the following questions, if switch 'S' is closed at t = 0. (Assume $e^{-1} = 0.37$, if required)



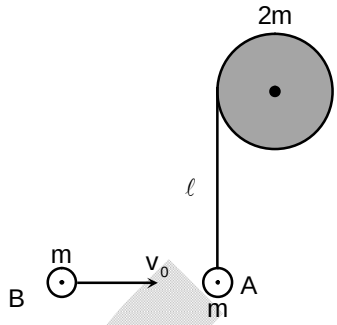
(A) Current in the branch of capacitor at t = 5/8 μs is (approximately) 6 A
(B) Current in the branch AB at t = 5/8 μs is (approximately) 1.7 A
(C) Potential difference between points A and C at t = 5/8 μs is (approximately) 2.5 V
(D) Potential difference between points A and C at t = 5/8 μs is (approximately) 6 V

3. A point mass m is attached to an ideal string, which in turns is wrapped on a vertical pole. Point mass is kept on a smooth horizontal surface and string is horizontal. A horizontal velocity is imparted to the point mass perpendicular to length of string.



(A) Speed of mass will become twice of initial value if length of string between the point mass and the pole becomes half of the original value.
(B) Kinetic energy of mass increases.
(C) Work done by tension on the point mass is zero.
(D) Energy of the mass remains constant.

4. In the adjacent figure, a uniform disc of mass 2m and radius $\ell/2$ is lying at rest on a smooth horizontal surface. A particle 'A' of mass m is connected to a light string of length ℓ, whose other end is attached to the circumference of the disc. Initially string is just taut and tangential to the disc, particle A is at rest. In the same horizontal plane another particle B of same mass m moving with velocity v_0 perpendicular to string collides elastically with A. Just after impact which of the following statements will be true



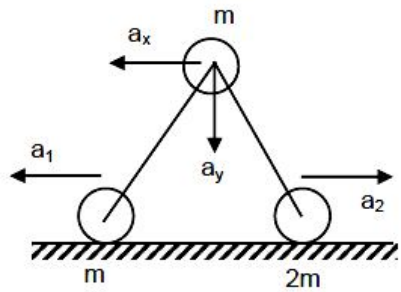
(A) Tension in the string is $\frac{2mv_0^2}{5\ell}$

(B) Acceleration of the centre of the disc is $\frac{v_0^2}{5\ell}$

(C) Initial radius of curvature of 'A' is $\frac{5\ell}{2}$

(D) Acceleration of the centre of the disc is $\frac{2v_0^2}{5\ell}$

5. Three masses are joined using pin joints (can move freely). Length of both rods are identical and system starts from rest. Accelerations of the balls are shown in figure. Angle between the rods at this moment is 90°. The horizontal surface is smooth. Choose the correct option(s).



(A) $2a_2 = a_1 + a_x$
(B) $2a_y = a_1 + a_2$
(C) $2a_x = a_1 - a_2$
(D) $3a_1 = 5a_2$

6. A particle of mass m is subjected to two forces

$\vec{F}_1 = f(r)\frac{\vec{r}}{|\vec{r}|}$ and $\vec{F}_2 = -\alpha\vec{v}$

Where f(r) is a real valued function depends on the distance of particle from origin, $\vec{r}$ is position vector of particle and α is positive constant. If at t = 0, $\vec{v}$ vector is in x-y plane and angular momentum of the particle about the origin is L_0. Choose the correct option(s).

(A) Plane of motion of particle does not change.

(B) Angular momentum of particle about the origin is given by $L_0 e^{-\frac{\alpha t}{m}}$

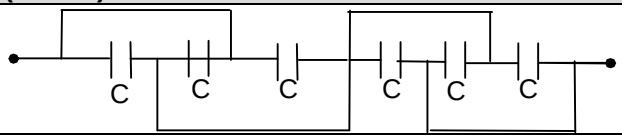
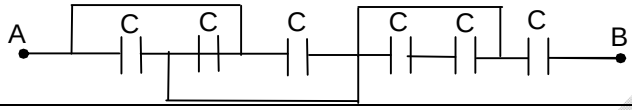
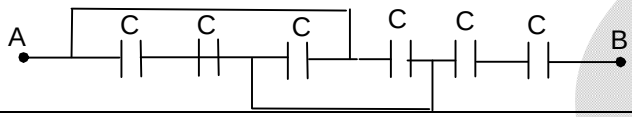
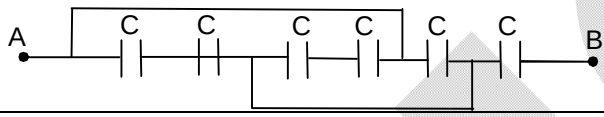
(C) Angular momentum of particle about the origin is remains constant.

(D) The value of angular momentum about the origin will depend on f(r)

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

7. Match List I with List II:

List I (Circuit)		List II (C_{eq})	
(I)		(P)	$\frac{3C}{4}$
(II)		(Q)	$\frac{5C}{12}$
(III)		(R)	$\frac{2C}{3}$
(IV)		(S)	$\frac{3C}{2}$

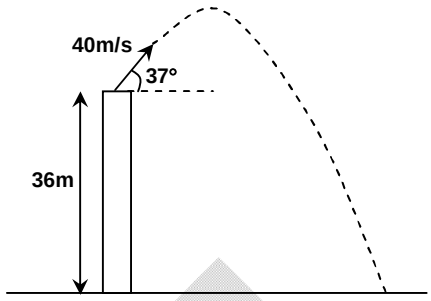
(A) I – S; II – Q; III – R; IV – P
(B) I – S; II – P; III – Q; IV – R
(C) I – S; II – P; III – R; IV – Q
(D) I – S; II – Q; III – P; IV – R

8. One mole of an ideal gas whose adiabatic exponent equals γ, undergoes a process $P = P_0 + \alpha/v$, where P_0 and α are positive constants. Its volume increases from v_1 to v_2.

List–I		List–II	
(I)	Heat capacity of gas as a function of its volume	(P)	$\frac{P_0(V_2 - V_1)}{\gamma - 1}$
(II)	Internal energy increment of a gas	(Q)	$\frac{P_0(v_2 - v_1)}{\gamma - 1} + \alpha \ln \frac{v_2}{v_1}$
(III)	The work performed by it	(R)	$P_0(v_2 - v_1) + \alpha \ln \frac{v_2}{v_1}$
(IV)	The amount of heat transferred	(S)	$\frac{\gamma R}{\gamma - 1} + \frac{\alpha R}{p_0 v}$

(A) I – S; II – P; III – R; IV – Q
(B) I – P; II – Q; III – R; IV – S
(C) I – P; II – S; III – R; IV – Q
(D) I – S; II – P; III – Q; IV – S

9. Consider a situation in which a projectile is fired at 40 m/s from the top of a tower in absence of air resistance t =0.



List–I		List–II	
(I)	Time at which particle moves at right angles with the initial direction	(P)	2.4 s
(II)	time at which kinetic energy is least	(Q)	6 s
(III)	Time at which it strikes the ground	(R)	20/3 s
(IV)	Time when the $\vec{v}$ make 45° with horizontal	(S)	5.6 s
		(T)	7.2 s

(A) (I) → (R); (II) → (P); (III) → (S); (IV) → (Q) (B) (I) → (P); (II) → (R); (III) → (Q); (IV) → (S)
(C) (I) → (P); (II) → (Q); (III) → (R); (IV) → (S) (D) (I) → (R); (II) → (P); (III) → (Q); (IV) → (S)

10. A neutron moving with kinetic energy K collides with a hydrogen atom at rest in ground state. Assume mass of neutron to be equal to that of hydrogen atom. Match the kinetic energy of neutron with the possible phenomena mentioned in column II. Energy of hydrogen in ground state is –13.6 ev

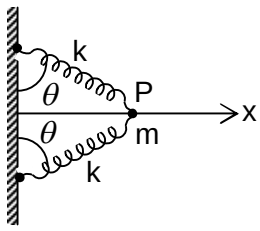
List– I		List–II	
(Kinetic energy of colliding neutron)			
I	K = 10.2 eV	(P)	Elastic collision may take place
II	K = 13.6 ev	(Q)	Inelastic collision may take place
III	K = 20.4 ev	(R)	Perfectly inelastic collision may take place
IV	K = 22.0 ev	(S)	Hydrogen atom may get ionised

(A) I-P; II-Q; III-R; IV-S (B) I-P; II-P; III-P, R; IV-P, Q
(C) I-P; II-P; III-P; IV-Q (D) I-P; II-P; III-R; IV-Q

Section – B (Maximum Marks: 24)

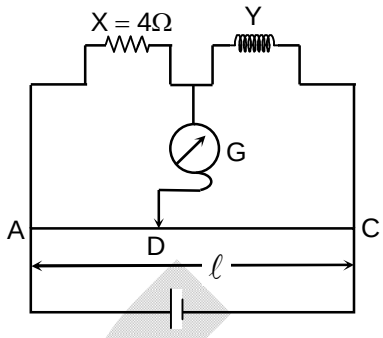
*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

11. Two identical springs each of stiffness k are welded at a point P. If a particle of mass m is welded at P. The period of oscillation of the particle in the direction of x is $\frac{\pi}{\sin\theta}\sqrt{\frac{Ym}{k}}$. Find Y

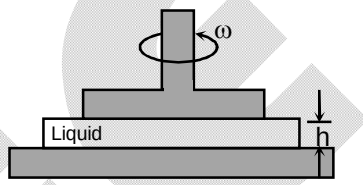


12. A body executing SHM with time period T and amplitude A. When it is at $\frac{A}{2}$ from equilibrium position moving toward right. It receives an impulse which doubles its velocity without changing direction. Time taken by the block to reach the right extreme position is $t = T\cos^{-1}\left(\frac{K}{\sqrt{13}}\right)$. Find K.

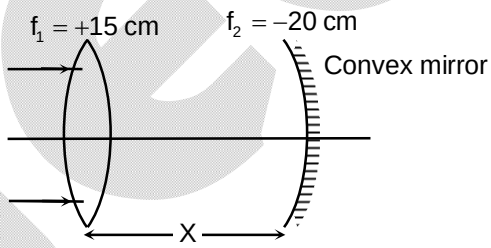
13. Figure shows a meter bridge wire AC has a uniform cross-section. The length of wire AC is 100 cm. X is a standard resistor of 4 Ω and Y is a coil. When Y is immersed in melting ice the null point is at 40 cm from point A. When the coil Y is heated to 100°C, a 12 Ω resistor has to be connected in parallel with Y in order to keep the bridge balanced at the same point.



14. A circular disc of a diameter d is slowly rotated in a liquid of large viscosity μ at a small distance h from a fixed surface. An expression for torque T necessary to maintain an angular velocity ω is $\frac{\mu\pi d^4\omega}{Xh}$. Find the value of X.



15. A set of parallel rays incident on the bi-convex lens. If after two refractions and one reflection final image formed at 15 cm left of lens, the value of $\frac{X}{9}$ is



16. Three SHM in the same direction in having same amplitude and same period are superposed. If each differ in phase from the next by 45°. Then, find the energy associated with resulting motion taking energy associated with each of given SHM as 1 J.

17. Two rods of different materials having coefficient of thermal expansion α_1, α_2 and Young's modulus Y_1, Y_2 are fixed between two massive walls. The rods are heated such that they undergo the same increase in temperature. If $\alpha_1 : \alpha_2 = 2 : 3$, then thermal stresses in rods are equal, then the ratio $Y_1 : Y_2$ is _____ .

18. A student determines a dimensionless quantity,

$$B = \frac{e^n}{Z\varepsilon_0 hc}.$$

Where, e = electric charge,
ε_0 = permittivity of vacuum,
h = Plank's constant
and c = speed of light.
Find the value of n.

Chemistry

PART – II

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

19. An isotone of $^{76}_{32}Ge$ is

(A) $^{77}_{32}Ge$ (B) $^{77}_{33}As$

(C) $^{77}_{34}Se$ (D) $^{78}_{34}Se$

20. For a compound AB crystallises in a rock salt structure with edge length a of the unit cell,

(A) the minimum distance between A and A is $a/\sqrt{2}$

(B) the minimum distance between B and B is $a/\sqrt{2}$

(C) the minimum distance between A and B is a/2

(D) the density of the crystal is given by $\rho = \frac{4}{a^3}\frac{M_{AB}}{N_A}$

21. Which of the following statements are not correct

(A) For an ideal binary liquid solution, the plot of $1/x_A$ versus $1/y_A$ (where x_A and y_A are the mole fractions of A in liquid and vapour phases, respectively) is linear.

(B) According to Raoult's law, the relative lowering of vapour pressure is equal to the mole fraction of solvent in the solution.

(C) The crysoscopic constant K_f is property of solvent.

(D) The value of van't Hoff factor for CH_3COOH in benzene will be 2

22. Which of the following statements are correct?

(A) Linear overlap of two atomic p-orbitals leads to a sigma bond

(B) The bond angle H – N – H in NH_3 is greater than the bond angle H – As – H in AsH_3.

(C) Anhydrous HCl is a bad conductor of electricity but aqueous HCl is a good conductor

(D) o-nitrophenol is steam volatile whereas p-nitrophenol is not.

23. Which of the following reactions give an alkyne?

(A) sodium fumarate $\xrightarrow[\text{electrolysis}]{\text{Kolbe's}}$

(B) $CH_3CBr_2CHBr_2 \xrightarrow[Zn/alcohol/\Delta]{}$

(C) $CH_3CH_2CHBr_2 \xrightarrow{NaNH_3/\Delta}$

(D) $CH_3CHBr - CHBr \xrightarrow{NaNH_2}$

24. Which of the following reactions give n-propyl alcohol as the major product?

(A) $CH_3 - CH = CH_2 + H_2O \xrightarrow{H^+}$

(B) $HCHO \xrightarrow[(ii)\,\text{hydrolysis}]{(i)\,CH_3CH_2MgBr}$

(C) $CH_3 - CH = CH_2 \xrightarrow{cold(aq)KMnO_4}$

(D) $CH_2 = CH - CHO + H_2 \xrightarrow{Ni}$

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match the entries in List-I for the spectral series of hydrogen atom with those given in List-II. The symbols n_1 and n_2 are the quantum level in the expression

List -I		List-II	
(I)	Lyman series	(P)	$n_1 = 2$ and $n_2 = 3, 4, 5,\ldots$
(II)	Balmer series	(Q)	$n_1 = 4$ and $n_2 = 5, 6, 7$
(III)	Paschen series	(R)	$n_1 = 5$ and $n_2 = 6, 7,\ldots$
(IV)	Pfund series	(S)	$n_1 = 1$ and $n_2 = 2, 3, 4,\ldots$
		(T)	$n_1 = 3$ and $n_2 = 4, 5, 6,\ldots.$

Which of the following is correct match
(A) I–S; (B) II–Q
(C) III–R (D) IV – T

26. Match the list -I and List- II

List -I		List-II	
(I)	Electrophilic substitution	(P)	$C_6H_5N_2Cl + C_6H_5OH \xrightarrow{NaOH}$
(II)	Nucleophilic substitution	(Q)	Toluene + NBS $\longrightarrow$
(III)	Free radical substitution	(R)	Dow's process
(IV)	Coupling reaction	(S)	Phenol $\longrightarrow$ picric acid
		(T)	$C_6H_5ONa + CH_3Br \longrightarrow$

Which of the following is correct match
(A) I–R, (B) II–P
(C) III-Q (D) IV–T

27. Match the list -I and List- II

List -I		List-II	
(I)	$2p\pi - 3p\pi$	(P)	$F_2C = O$
(II)	$3p\pi - 3d\pi$	(Q)	$F_3P = O$
(III)	$2p\pi - 3d\pi$	(R)	$F_3P = S$
(IV)	$2p\pi - 2p\pi$	(S)	$F_2S = O$
		(T)	$F_2C = S$

Which of the following is correct match
(A) I–R, (B) II–T
(C) III-Q (D) IV–P

28. List-II give the values of γ (- $C_{p,m}/C_{V,m}$) at high temperature of gases which are shown in column I. Identify the gas with the corresponding γ values.

List -I		List-II	
(I)	Monatomic	(P)	1.154
(II)	Diatomic	(Q)	1.286
(III)	Linear Triatomic	(R)	1.167
(IV)	Nonlinear Triatomic	(S)	1.407
		(T)	1.667

Which of following is correct match
(A) I–Q (B) II–T
(C) III-P (D) IV–R

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

29. The ionization potential of hydrogen atom is 13.6 eV. The energy in electron volt required to remove an electron in the n = 2 state of the hydrogen atom is

30. The density of a gas is 1.5 g L^{-1} at STP. Its density at 27 ºC and 1000 Torr would be

31. Ammonium chloride crystallizes in a body-centred cubic lattice with edge length of 387 pm. If the size of NH_4^+ is 154 pm, the size of Cl^- ion would be

32. A container of volume 'V' litre contains an equilibrium mixture which consists of 2 mol each of PCl_5, PCl_3 and Cl_2. The equilibrium pressure is 3 atm and temperature is 'T' kelvin. A certain amount of Cl_2 gas is now introduced into the container keeping the temperature and presence constant, until the equilibrium volume becomes '2V' litre. How many moles of Cl_2 were added?

33. The conductivity of a saturated solution of silver oxalate is 4.5×10^{-5} S cm^{-1}. If its $K_{sp} = 1.35 \times 10^{-11} M^3$, the molar conductivity of the saturated solution would be

34. How many grams of potassium perchlorate ($KClO_4$) must be mixed with 108 g of water in order to obtain a solution, which has a mole fraction of $KClO_4$ equal to $\frac{1}{7}$?

35. Calculate effective nuclear charge on last electron of Cl. (using Slater rule)

36. The number of isomers possible for complex $K_2[PtClBr_2SCN]$:

Mathematics

PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

37. A point $P\left(\sqrt{3},1\right)$ moves on the circle $x^2+y^2=4$ and after covering a quarter of the circle leaves it tangentially. The equation of a line along which the point moves after leaving the circle is

(A) $y=\sqrt{3}x+4$ (B) $\sqrt{3}y=x+4$

(C) $\sqrt{3}y=x-4$ (D) $y=\sqrt{3}x-4$

38. A chord PP' of a parabola cuts the axis of the parabola at O. The feet of the perpendiculars from P and P' on the axis are M and M' respectively. If V is the vertex then VM, VO, VM' are in

(A) AP (B) GP

(C) HP (D) none of these

39. α, β and γ are three angles given by $\alpha=2\tan^{-1}\left(\sqrt{2}-1\right),\beta$

$=3\sin^{-1}\frac{1}{\sqrt{2}}+\sin^{-1}\left(-\frac{1}{2}\right)$ *and* $\gamma=\cos^{-1}\frac{1}{3}$. Then

(A) $\alpha>\beta$ (B) $\beta>\gamma$

(C) $\alpha<\gamma$ (D) none of these

40. $\int_4^8\left(\sqrt{\sqrt{16x-64}+x}+\sqrt{x-\sqrt{16x-64}}\right)dx$

(A) $2\int_4^6\left(\sqrt{\sqrt{16x-64}+x}+\sqrt{x-\sqrt{16x-64}}\right)dx$ (B) $2\int_8^{10}\left(\sqrt{\sqrt{16x-64}+x}+\sqrt{x-\sqrt{16x-64}}\right)dx$

(C) $2\int_7^9\left(\sqrt{\sqrt{16x-64}+x}+\sqrt{x-\sqrt{16x-64}}\right)dx$ (D) 16

41. If M is the skew symmetric matrix of order n × n, then

(A) det (M – I) = det (M + I) (B) det (M – I) = – det (M + I) if n is odd

(C) det (M – I) = det (M + I) if n is even (D) det (I – M) $(I + M)^{-1}$) = 1, det (I + M) $\neq 0$

42. If the unit vectors $\vec{a}$ and $\vec{b}$ are inclined at an angle 2θ such that $\left|\vec{a}-\vec{b}\right|<1$ and $0\le\theta\le\pi$, then θ lies in the interval

(A) [0, π/6) (B) (5π/6, π]

(C) (π/6, π/2] (D) [π/2, 5π/6)

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

43. Match the List– I & List–II

List–I		List–II	
I	A straight line with negative slope passing through (1, 4) meets the coordinate axes at A and B. The minimum length of OA + OB, O being the origin, is	P	$5\sqrt{2}$
II	If the point P is symmetric to the point Q(4, –1) with respect to the bisector of the first quadrant, then the length of PQ is	Q	$3\sqrt{2}$
III	On the portion of the straight line x + y = 2 between the axis a square is constructed away from the origin, with this portion as one of its sides. If d denotes the perpendicular distance of a side of this square from the origin then the maximum value of d is	R	9/2
IV	If the parametric equation of a line is given $x = 4 + \lambda/\sqrt{2}$ and $y = -1 + \sqrt{2}\lambda$, where λ is a parameter, then the intercept made by the line on the x-axis is	S	9

(A) I $\rightarrow$ R; II $\rightarrow$P; III $\rightarrow$ Q; IV $\rightarrow$ S
(B) I $\rightarrow$ S; II $\rightarrow$P; III $\rightarrow$ Q; IV $\rightarrow$ R
(C) I $\rightarrow$ S; II $\rightarrow$R; III $\rightarrow$ Q; IV $\rightarrow$ P
(D) I $\rightarrow$ P; II $\rightarrow$S; III $\rightarrow$ Q; IV $\rightarrow$ R

44. Match the List–I & List–II

List–I		List–II	
I	The smallest integer greater than $\frac{1}{\log_3 \pi}+\frac{1}{\log_4 \pi}$ is	P	10
II	Let $3^a = 4, 4^b = 5, 5^c = 6, 6^d = 7, 7^e = 8$, and $8^f = 9$. Then the value of the product (abcdef) is	Q	3
III	Characteristic of the logarithm of 2008 to the base 2 is	R	1
IV	If $\log_2\left(\log_2\left(\log_3 x\right)\right) = \log_2\left(\log_3\left(\log_2 y\right)\right) = 0$, then the value of (x – y) is	S	2

(A) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ Q
(B) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ R
(C) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ R
(D) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ S

45. Match the following List-I with List–II

List–I		List–II	
I	$\lim\limits_{x\to 0}\frac{1-\cos 2x}{e^{x^2}-e^x+x}$ equals to	P	1
II	If the value of $\lim\limits_{x\to 0^+}\left(\frac{(3/x)+1}{(3/x)-1}\right)^{1/x}$ can be expressed in the form of $e^{p/q}$, where p and q are relative prime, then (p + q) is equal to	Q	2
III	$\lim\limits_{x\to 0}\frac{\tan^3 x-\tan x^3}{x^5}$ equals to	R	4
IV	$\lim\limits_{x\to 0}\frac{x+2\sin x}{\sqrt{x^2+2\sin x+1}-\sqrt{\sin^2 x-x+1}}$ equals to	S	5

(A) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ Q
(B) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ S; IV $\rightarrow$ P
(C) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ Q
(D) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ R

46. Let Π be the plane parallel to y-axis and containing the points (1, 0, 1) and (3, 2, -1). A ≡ (4, 0, 0) and B ≡ (6, 0, –2) are two points and P ≡ (x_0, y_0, z_0) is a variable point on the plane Π = 0.
Match List – I with List – II and select the correct answer using the code given below the list.

List - I		List - I	
I	If the equation of the plane $\Pi = 0$ is x + ay + bz = c, then |a + b + c| is	P	16
II	If (PA + PB) is minimum, then $\left\|4x_0 + y_0 + 2z_0\right\|$ is	Q	12
III	If |PA + PB| ∈ $[0, \sqrt{N})$, then N is	R	3
IV	If the reflection of the line AB in the plane $\Pi = 0$ is $\frac{x-2}{1} = \frac{y-\alpha}{0} = \frac{z+\beta}{-1}$, then $(\alpha^4 + \beta^4)$	S	8
		T	10

(A) I → R, II → Q, III → P, IV → S (B) I → Q, II → R, III → S, IV → P
(C) I → R, II → Q, III → S, IV → P (D) I → R, II → S, III → Q, IV → P

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

47. In the coordinate plane, the region M consists of all the points (x, y) satisfying the inequalities $y \geq 0, y \leq x\,y$ and $y \leq 2 - x$ simultaneously. The region N, which varies with the parameter t, consists of all the points (x, y) satisfying the inequalities $t \leq x \leq t+1$ and $0 \leq t \leq 1$ simultaneously. The area of $M \cap N$ is $-k_1t^2 + k_2t + \frac{1}{2}$, then $k_1 + k_2$ is _____ .

48. The slope of the tangent to the curve represented by $x = t^2 + 3t - 8$ and $y = 2t^2 - 2t - 5$ at the point $M(2, 1)$ is $\left(\frac{\lambda}{7}\right)$, then value of λ is ?

49. Let $(1+3x+4x^2) = a_0 + a_1x + a_2x^2 + .. + a_{20}x^{20}$. Then value of a_3 is –

50. Find the value of $f\left(\frac{\pi}{6}\right)$, where $f(\theta) = \begin{vmatrix} \cos^2\theta & \cos\theta\sin\theta & -\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix}$

51. If α, β, γ are such that $\alpha + \beta + \gamma = 4$, $\alpha^2 + \beta^2 + \gamma^2 = 6$, $\alpha^3 + \beta^3 + \gamma^3 = 8$, then the value of $\left[\alpha^4 + \beta^4 + \gamma^4\right]$ must be equal to (where [.] denotes the greatest integer function)

52. Find the Coefficient of x^{103} in $(1 + x + x^2 + x^3 + x^4)^{199}(x - 1)^{201}$.

53. If $\left(1+\frac{1}{x}\right)^{x+1} = \left(1+\frac{1}{2021}\right)^{2021}$, then |x| is equal to

54. If two events A and B are such that $P(\bar{A}) = 0.3$, P(B) = 0.4, $P(A \cap \bar{B}) = 0.5$, then the value of $P(B/(A \cup \bar{B}))$ is